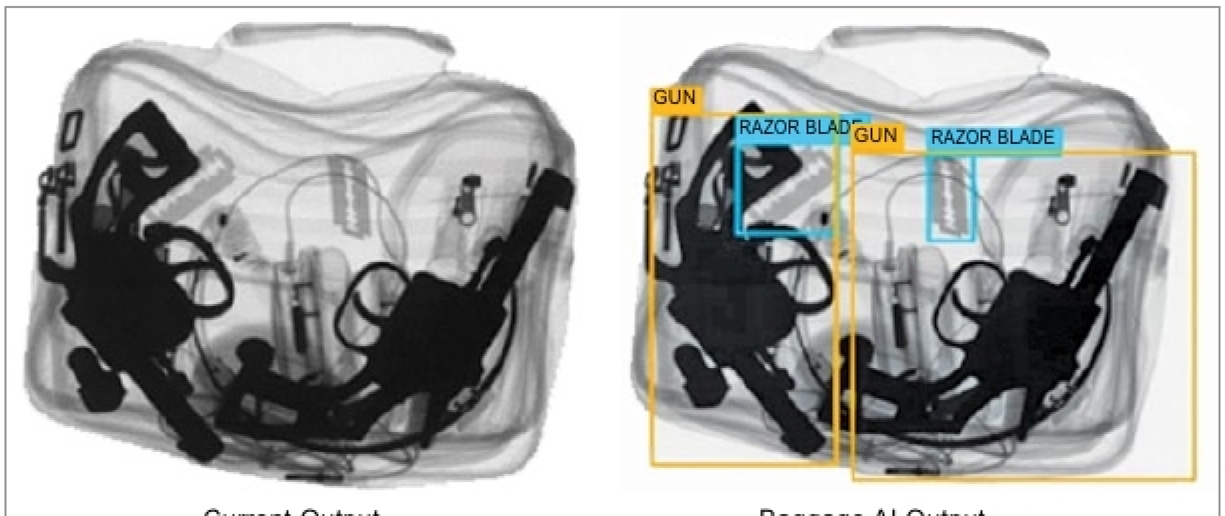

Current Output Baggage AI Output

sity of objects in the bag to construct an image showing the position of objects and their density. That is how existing technology differentiates flour from cocaine, talcum powder from explosive ones and regular plastic from a bomb casing.

Manual detection of banned items in x-ray images is an intricate task and is highly prone to human error due to lethargy. Baggage handling and security are major operational challenges, and sometimes contribute to flight delays as well. To mitigate these challenges, a high-performance intelligent baggage scanner system is needed.

BaggageAI has been developed targeting airports for the most frequent common items, which are contraband like power bank, coconut, lighter and e-cigarette. It has been tested by Delhi International Airport Ltd, India. The solution has also been selected by Airport Authority of India for pilot and deployment to improve efficiency of security personnel.

It is a sophisticated high-performance artificial intelligence (AI)-based system for automatic object detection of banned items in baggage through x-ray images. It looks at each image like a human eye, outlines banned objects and sends alerts to the screen. This is very different from density-based object detection. Existing scanners work on manual detection, but this solution automates the process of finding and locating prohibited items in the x-ray image of the scanned baggage.

BaggageAI can scan thirty images per second, and detect banned objects from 2D black-and-white x-ray images, coloured x-ray images as well as 3D CT scan images. It does not need any extra hardware and can be integrated with the existing 2D or 3D scanning machines to improve their performance. It helps detect threats with high accuracy and efficiency, and reduces false alarms, saving invaluable time.

It has a high throughput of up to thirty images per second, which helps airports handle more passengers and flights per day, resulting in higher revenues and profits. This boosts passenger security significantly by removing human errors in the baggage screening process. It can be effectively deployed at places with high passenger traffic such as airports, metros, malls and railway stations where there is a security risk.

"BaggageAI has deep learning algorithms at its core so a significant time was taken to develop it. Getting the right data and testing it in a real-time situation were some of the major challenges faced by the team. It is being tested by multiple vendors across Europe and North America. The current system is under constant improvement and will be improved to be used across verticals like vehicle scanning, cargo scanning and others," says Himanshu Arora, co-founder, Dimensionless Technologies Pvt Ltd.

—Deepshikha Shukla

4 LEAP MOTION CONTROLLER

In future, there will be two worlds, one where you interact naturally and in 3D, and another where you will need controllers to act. Tracking and mid-air haptics are transforming interactions of the virtual world. These can revolutionise 3D automotive controls, bringing the sense of touch into virtual reality (VR) and augmented reality (AR).

Ultraleap was formed when Leap Motion and Ultrahaptics came together in 2019. They united advanced hand tracking with haptic technology that creates sensation of touch in mid-air. Ultraleap's optical hand tracking module captures movements of hands with unparalleled accuracy and near-zero latency.

It is a robust and reliable skeletal model that has motion-to-photon latency below the human perception threshold. It makes human interaction in the digital world natural and effortless. It utilises two image sensors plus infrared (IR) LEDs. The LEDs illuminate hands with IR pulses over hundred times a second. Every time the LEDs illuminate, sensors send data back to the computer to track hands. It is developed for public use with CE, FCC and CAN ICES-3 certifications.